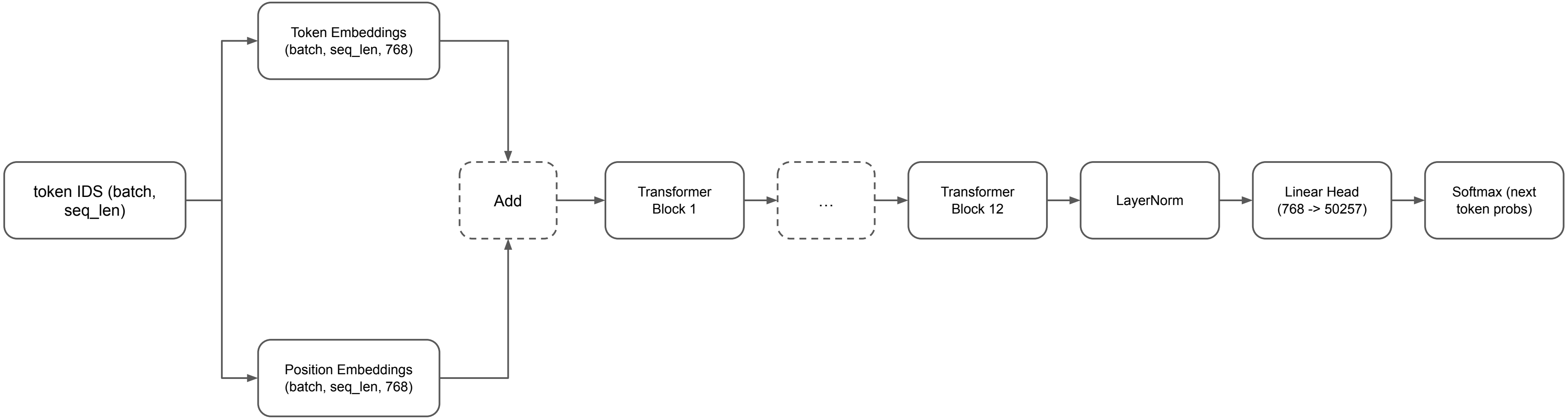
Pretrain GPT from scratch

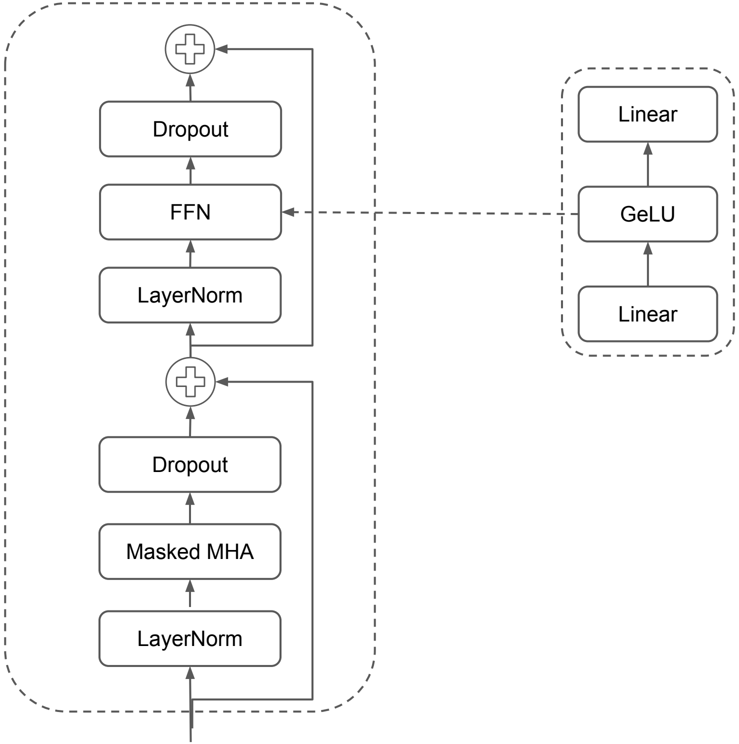
AI Engineering From scratch #10

GPT-2 Architecture



GPT-2 Architecture

Transformer block architecture



KV cache

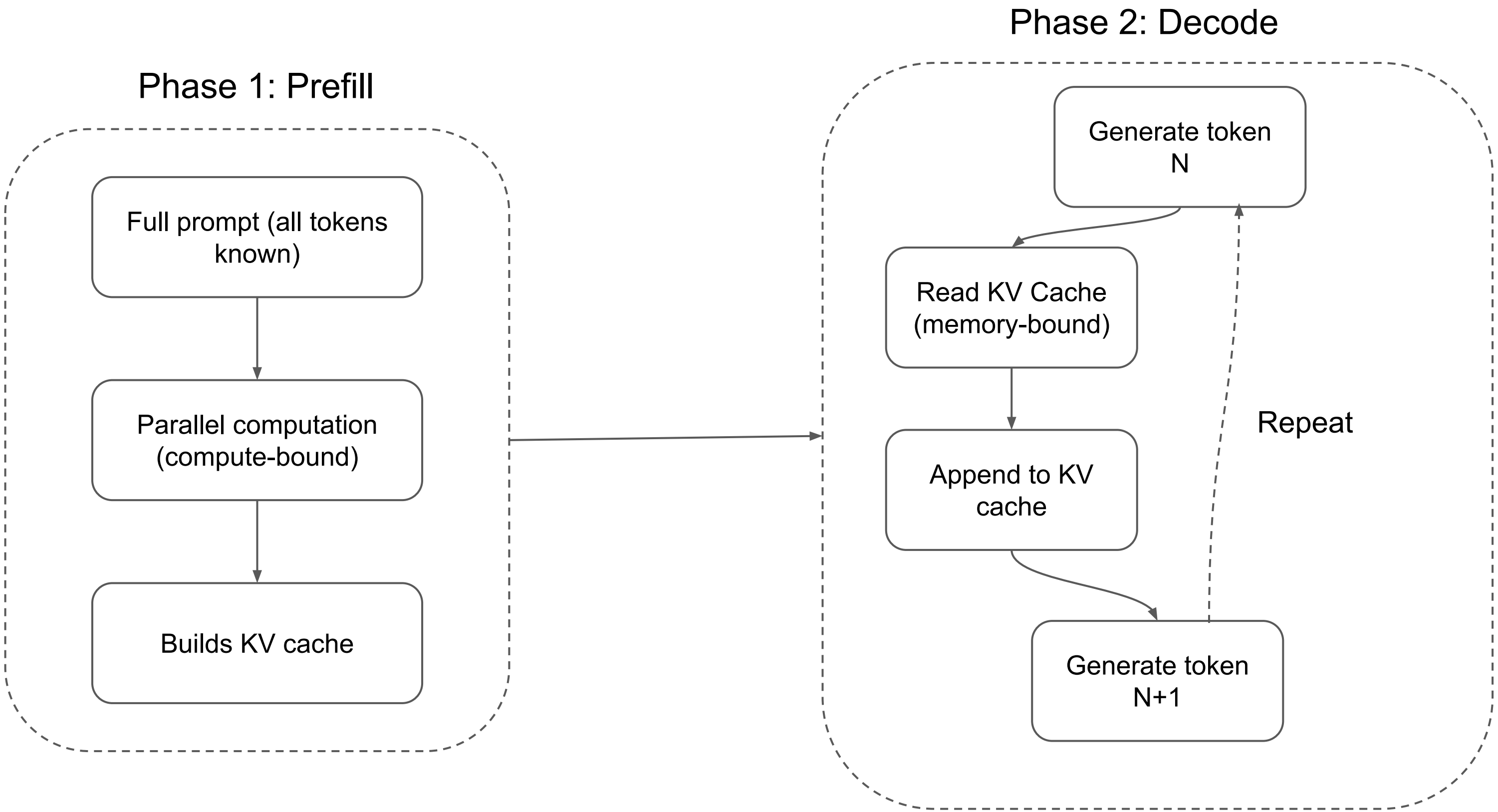
- During training, you process the entire sequence at once. During inference, you generate one token at a time. Without optimization, generating token N requires recomputing attention for all N-1 previous tokens. That is O(N^2) per generated token.
- KV Cache solves this:
    - After computing K and V for each token, store them.
    - When generating token N+1, you only need to compute Q for the new token and look up the cached K and V from all previous tokens.
    - This reduces per-token cost from O(N) to O(1) for the K and V computation.
    - → Avoid redundant matrix multiplications on the input.

Prefill vs Decode: Two phases of inference

When you send a prompt to an LLM, inference happens in two distinct phases:

- Prefill:
    - Processes your entire prompt in parallel.
    - All tokens are know, so the model can compute attention for all positions simultaneously.
- Decode:
    - Generates tokens one at a time.
    - Each new token depends on all previous tokens.

Prefill vs Decode: Two phases of inference



The Training Loop

One training step:

- Forward pass:
  - Run the batch through all 12 blocks. Get logits (pre-softmax scores) for each position.
- Compute loss:
  - Cross-entropy between logits and target tokens (the input shifted by one position).
- Backward pass:
  - Compute gradients for all 124M parameters using backpropagation.
- Optimizer step:
  - Update weights. GPT-2 uses Adam with learning rate warmup and cosine decay.